Cambridge Assessment International Education

Cambridge IGCSE™

CANDIDATE NAME			
CENTRE NUMBER		CANDIDATE NUMBER	

PHYSICS **0625/42**

Paper 4 Theory (Extended) **Practice Paper**

1 hour 15 minutes

You must answer on the question paper.

No additional materials are needed.

INSTRUCTIONS

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.
- You should show all your working and use appropriate units.
- Take the weight of 1.0 kg to be 10 N (acceleration of free fall = 10 m/s^2).

INFORMATION

- The total mark for this paper is 80.
- The number of marks for each question or part question is shown in brackets [ ].

This document has 8 pages. Any blank pages are indicated.

1 Fig. 1.1 shows an electrically powered bicycle.

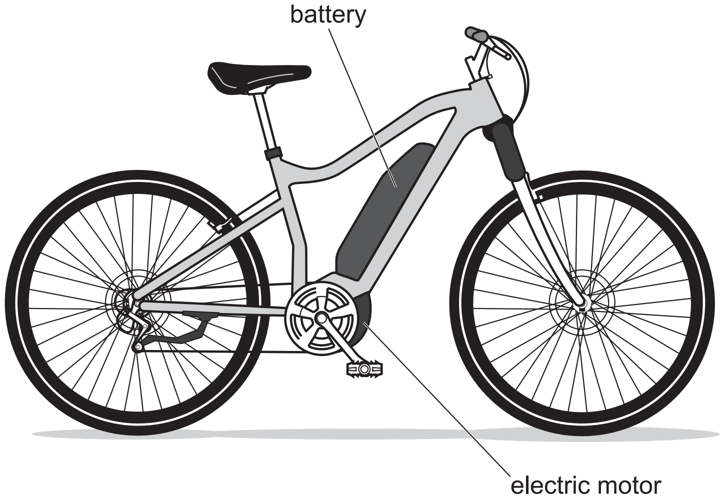


Fig. 1.1

The battery can deliver a power of 600 W for 60 min.

(a) **(i)** Calculate the energy, in joules, stored in the battery when fully charged.

energy = ... J [3]

(ii) State the form of energy stored by the battery.

.... [1]

(b) The bicycle has a motor with an electrical input power of 250 W.

Calculate the time for which the battery can power the bicycle.

time = ... s [2]

(c) State two environmental benefits of using an electrically powered bicycle instead of a small motorcycle.

1. ....

2. ....

[2]

[Total: 8]

2 Fig. 2.1 shows two objects connected by a cord passing over a pulley.

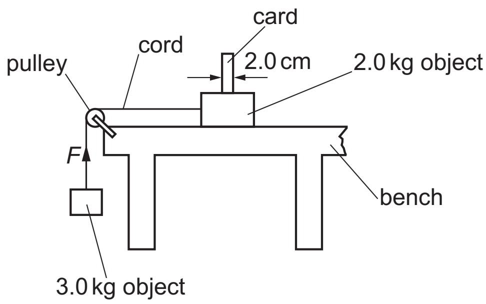


Fig. 2.1

The 2.0 kg object is released from rest and accelerates at 4.0 m/s^2.

(a) Calculate the resultant force acting on the 2.0 kg object.

force = ... N [2]

(b) Calculate the upward force exerted by the cord on the 3.0 kg object.

force = ... N [3]

(c) **(i)** The objects have a constant acceleration.

Show that the speed of the objects 0.80 s after release is 3.2 m/s.

[2]

(ii) A card of width 2.0 cm passes through a beam of light at this speed. Calculate the time taken for the card to pass through the beam.

time = ... s [2]

[Total: 9]

3 **(a)** Fig. 3.1 shows water in a river moving parallel to the river bank at 4.0m/s and a canoe travelling in the river.

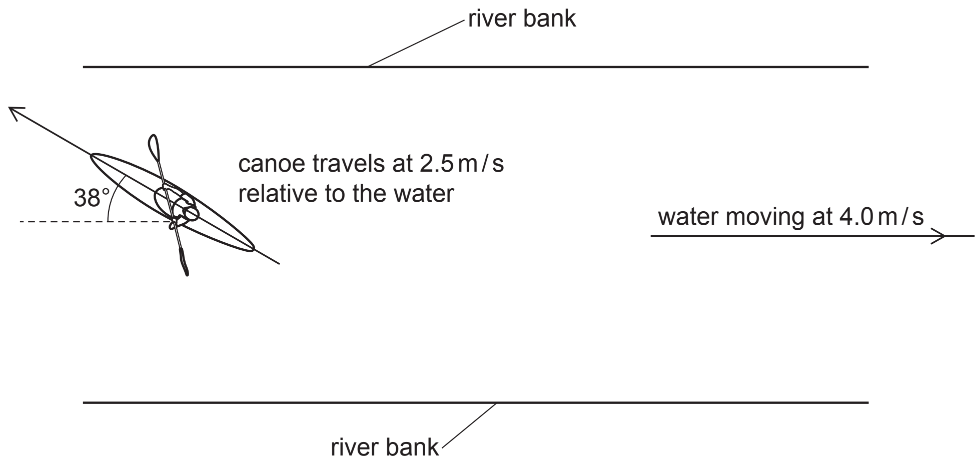


Fig. 3.1

The canoe travels at 2.5m/s relative to the water and heads at an angle of 38° to the river bank.

Draw a scale diagram to determine the canoe's resultant velocity and state the scale you used.

scale ....

magnitude of resultant velocity ....

direction of resultant velocity (angle from the river bank) .... [4]

(b) Explain why the resultant velocity is not in the same direction as the canoe points.

....

.... [2]

[Total: 8]

4 A student investigates the resistance of a wire.

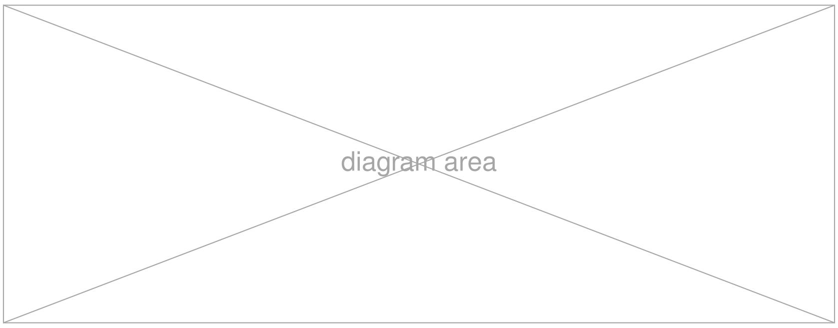


Fig. 4.1

(a) Complete the table headings for the measurements needed in this experiment.

Length of wire / cm		
20		
40		
60		
80		

[2]

(b) Describe how the student can use the measurements to determine the resistance of the wire.

[4]

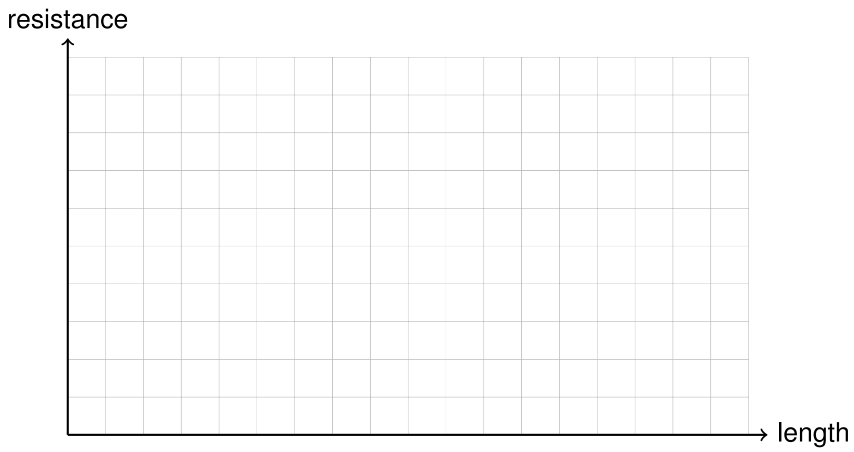
(c) State two precautions that improve the reliability of the results.

1. ....

2. ....

[2]

(d) On the grid, sketch the expected shape of a graph of resistance against length for a uniform wire.



[2]

[Total: 10]

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5 This page shows the structure for adding another long question. Replace this text with your own question stem.

(a) Write the first part of the question here.

answer = ... unit [2]

(b) Write the second part of the question here.

[4]

[Total: 6]